

Unit 202 — Instructional Visual Review Package

Generated 2026-08-23T01:55:04.588Z

31 canonical instructional-visual variants, one per page, at a realistic mobile viewport width (390px).

This package is for REAL, human/external pixel-level review. No image in this package has been visually approved by this repository's tooling. Any "SIMULATED SEMANTIC" result referenced elsewhere in this project's reports comes from a deterministic mock pipeline that proves the QA architecture works end-to-end -- it is not a substitute for the review this package exists to support.

Each page shows: the actual rendered instructional SVG at mobile scale, its governed instructional purpose, which assessment-obligation/assertion/capability it is linked to, its TEACHING / ASSESSMENT / BOTH state, and the semantic contract's must-show / must-not-show requirements it is being checked against.

Show a sine AC waveform with correctly-positioned peak/RMS/period reference marks, teaching the $\text{RMS} \sim 0.707 \times \text{peak}$ relationship and the zero full-cycle average of symmetric AC.

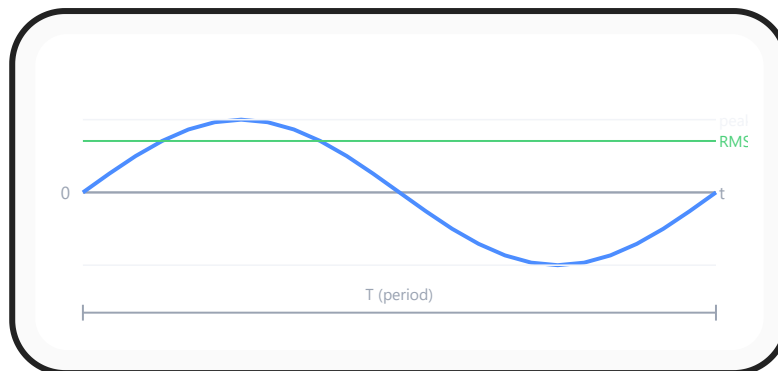


Diagram blueprint

`graph.waveform_sine (waveform)`

Visual semantic contract

`visual-contract.ac-waveform-sine`

Canonical variant id

`visual-contract.ac-waveform-sine@1::cycles_shown=1,mode=both,show_peak_line=true,show_period_marker=true,show_rms_line=true::both`

Linked assertion family

`electrical.ac_dc_waveforms`

Linked assertion identifiers

`EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001`

Linked capability

`cap.waveform.identify_characteristic, cap.waveform.calculate_rms_peak`

Representation role

`essential`**MUST SHOW**

- the sine curve itself over `cycles_shown` full periods
- the zero/reference axis, which every other reference line is measured from

MUST NOT SHOW

- an RMS line coincident with the peak line
- a non-zero horizontal 'average' reference line for a full symmetric cycle (the real signed average is zero -- `EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001`)
- any numeric peak/RMS/period value embedded in the artwork itself (values belong in question/lesson prompt text, per the symbolic-only convention every diagram in this folder follows)

Show a sine AC waveform with correctly-positioned peak/RMS/period reference marks, teaching the $\text{RMS} \sim 0.707 \times \text{peak}$ relationship and the zero full-cycle average of symmetric AC.

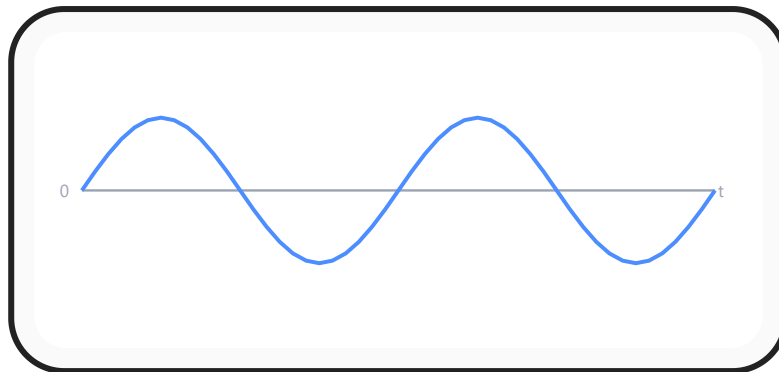


Diagram blueprint

`graph.waveform_sine (waveform)`

Visual semantic contract

`visual-contract.ac-waveform-sine`

Canonical variant id

`visual-contract.ac-waveform-sine@1::cycles_shown=2,mode=both,show_peak_line=false,show_period_marker=false,show_rms_line=false::both`

Linked assertion family

`electrical.ac_dc_waveforms`

Linked assertion identifiers

EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001

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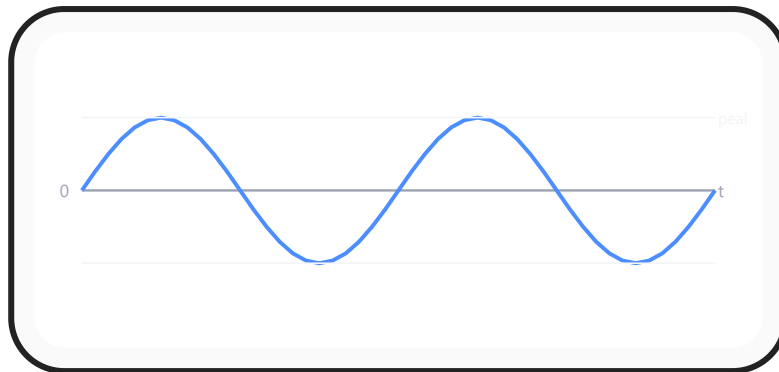


Diagram blueprint

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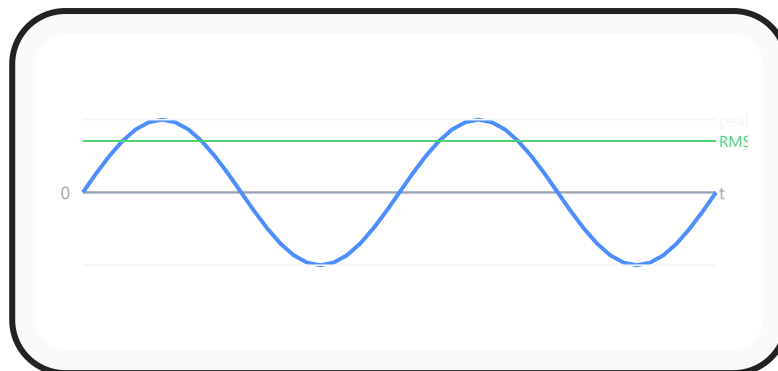


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

`graph.waveform_sine (waveform)`

`visual-contract.ac-waveform-sine`

`visual-contract.ac-waveform-sine@1::cycles_shown=2,mode=both,show_peak_line=true,show_period_marker=false,show_rms_line=true::both`

`electrical.ac_dc_waveforms`

EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001

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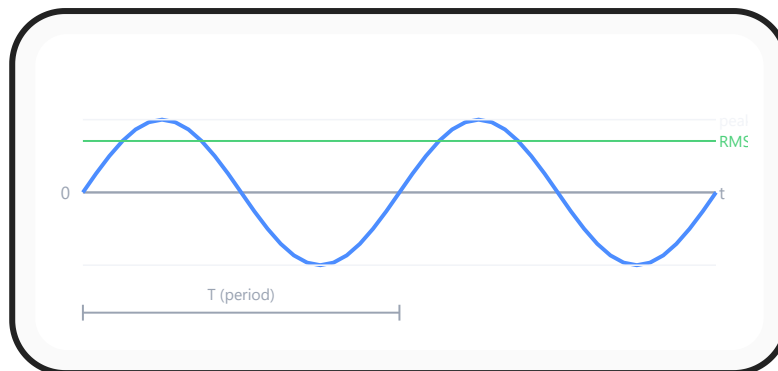


Diagram blueprint

`graph.waveform_sine (waveform)`

Visual semantic contract

`visual-contract.ac-waveform-sine`

Canonical variant id

`visual-contract.ac-waveform-sine@1::cycles_shown=2,mode=both,show_peak_line=true,show_period_marker=true,show_rms_line=true::both`

Linked assertion family

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Linked assertion identifiers

`EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001`

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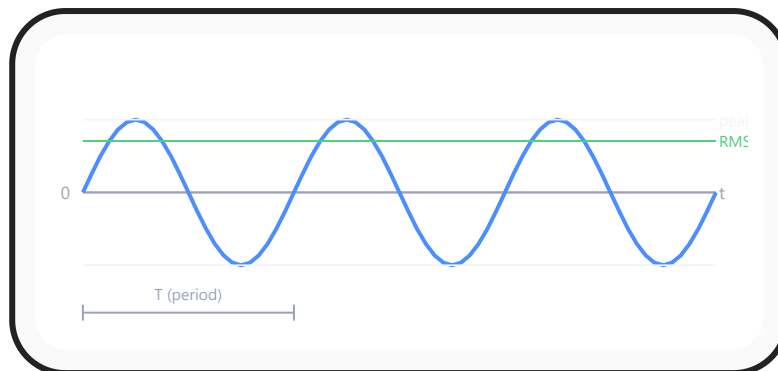


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

`graph.waveform_sine (waveform)``visual-contract.ac-waveform-sine``visual-contract.ac-waveform-sine@1::cycles_shown=3,mode=both,show_peak_line=true,show_period_marker=true,show_rms_line=true::both``electrical.ac_dc_waveforms`

EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001

`cap.waveform.identify_characteristic, cap.waveform.calculate_rms_peak``essential`**MUST SHOW**

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Teach correct voltmeter-parallel / ammeter-series connection, and that ohmmeter use requires a de-energised, isolated component.

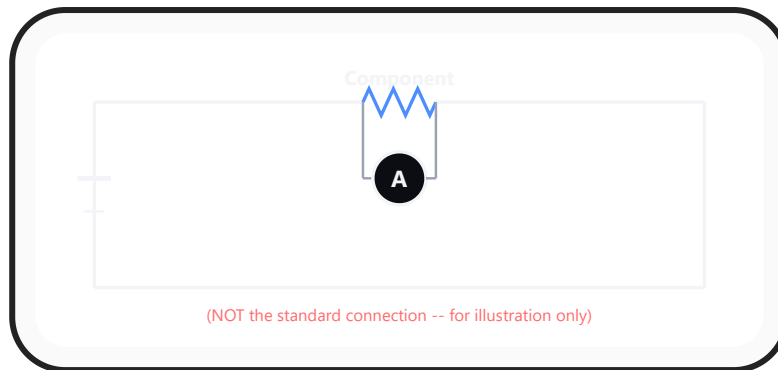


Diagram blueprint	<code>instrument.measurement_connection (instrument_connection)</code>
Visual semantic contract	<code>visual-contract.instrument-measurement-connection</code>
Canonical variant id	<code>visual-contract.instrument-measurement-connection@1::connection_style=parallel,instrument_type=ammeter,mode=both::both</code>
Linked assertion family	<code>electrical.instrumentation</code>
Linked assertion identifiers	<code>EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001</code>
Linked capability	<code>cap.instrumentation.recognise_connection</code>
Representation role	<code>essential</code>

MUST SHOW

- the instrument symbol (circle containing V/A/Ω) at the correct topological position for the requested `connection_style`
- the component/path being measured
- an explicit on-diagram caption stating whether the shown combination matches the standard, correct connection method for that instrument (never silently implying a miswiring is standard practice)

MUST NOT SHOW

- a voltmeter spliced into the current path (the main path must remain unbroken whenever a voltmeter is shown)
- an ammeter drawn as a side branch (it must always be spliced into the current path)
- for instrument_type 'ohmmeter', any energised-source symbol in a closed loop with the meter -- ohmmeter use requires an isolated, de-energised component

Teach correct voltmeter-parallel / ammeter-series connection, and that ohmmeter use requires a de-energised, isolated component.

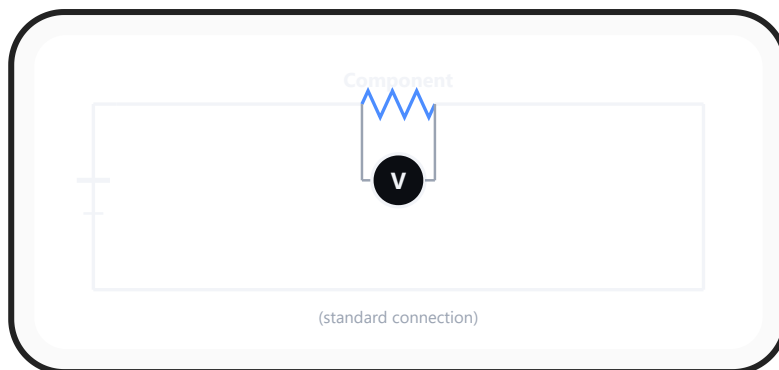


Diagram blueprint

instrument.measurement_connection (instrument_connection)

Visual semantic contract

visual-contract.instrument-measurement-connection

Canonical variant id

visual-contract.instrument-measurement-connection@1::connection_style=parallel,instrument_type=voltmeter,mode=both::both

Linked assertion family

electrical.instrumentation

Linked assertion identifiers

EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001

Linked capability

cap.instrumentation.recognise_connection

Representation role

essential

MUST SHOW

- the instrument symbol (circle containing V/A/Omega) at the correct topological position for the requested connection_style
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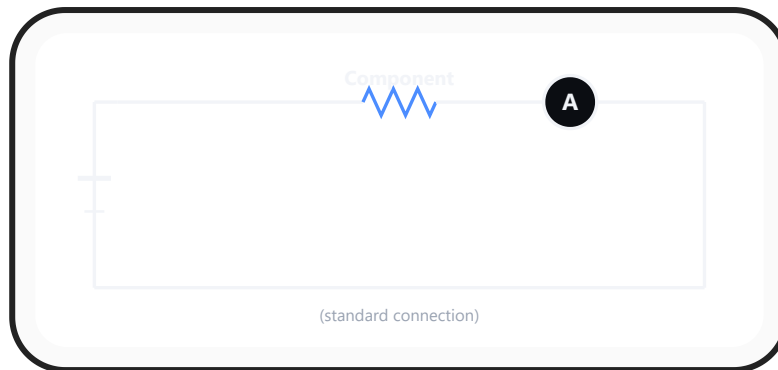


Diagram blueprint

instrument.measurement_connection (instrument_connection)

Visual semantic contract

visual-contract.instrument-measurement-connection

Canonical variant id

visual-contract.instrument-measurement-connection@1::connection_style=series,instrument_type=ammeter,mode=both::both

Linked assertion family

electrical.instrumentation

Linked assertion identifiers

EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001

Linked capability

cap.instrumentation.recognise_connection

Representation role

essential

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Teach correct voltmeter-parallel / ammeter-series connection, and that ohmmeter use requires a de-energised, isolated component.

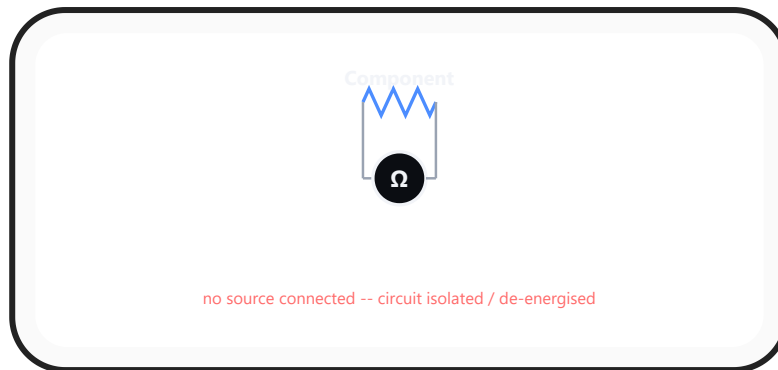


Diagram blueprint	<code>instrument.measurement_connection (instrument_connection)</code>
Visual semantic contract	<code>visual-contract.instrument-measurement-connection</code>
Canonical variant id	<code>visual-contract.instrument-measurement-connection@1::connection_style=series,instrument_type=ohmmeter,mode=both::both</code>
Linked assertion family	<code>electrical.instrumentation</code>
Linked assertion identifiers	<code>EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001</code>
Linked capability	<code>cap.instrumentation.recognise_connection</code>
Representation role	<code>essential</code>

MUST SHOW

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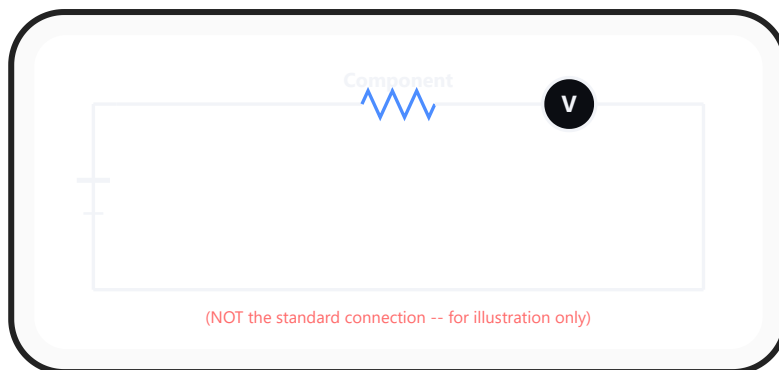


Diagram blueprint

instrument.measurement_connection (instrument_connection)

Visual semantic contract

visual-contract.instrument-measurement-connection

Canonical variant id

visual-contract.instrument-measurement-connection@1::connection_style=series,instrument_type=voltmeter,mode=both::both

Linked assertion family

electrical.instrumentation

Linked assertion identifiers

EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001

Linked capability

cap.instrumentation.recognise_connection

Representation role

essential

MUST SHOW

- the instrument symbol (circle containing V/A/Omega) at the correct topological position for the requested connection_style
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Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

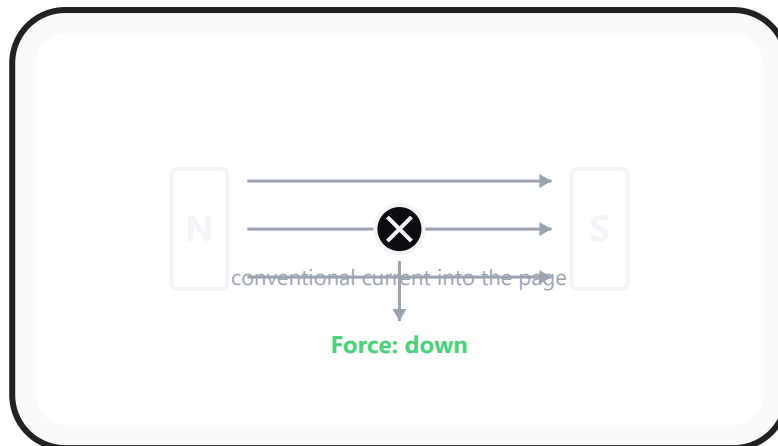


Diagram blueprint

Visual semantic contract

Canonical variant id

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=into_page,force_direction=down,mode=teaching,pole_labels=N_S_horizontal,show_force_arrow=true::teaching

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

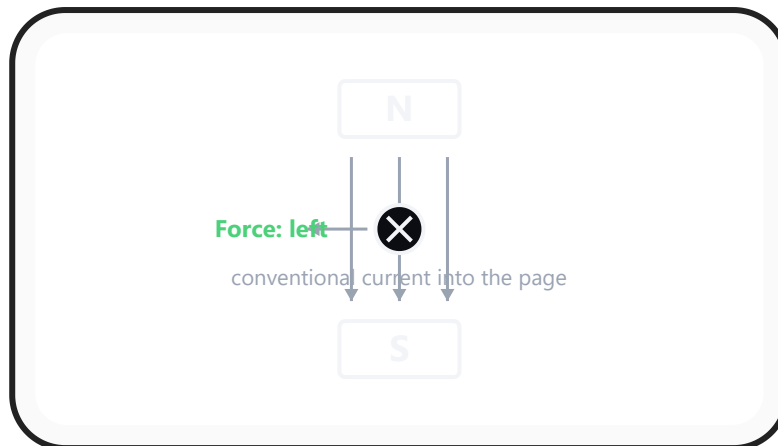


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=into_page,force_direction=left,mode=teaching,pole_labels=N_S_vertical,show_force_arrow=true::teaching

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

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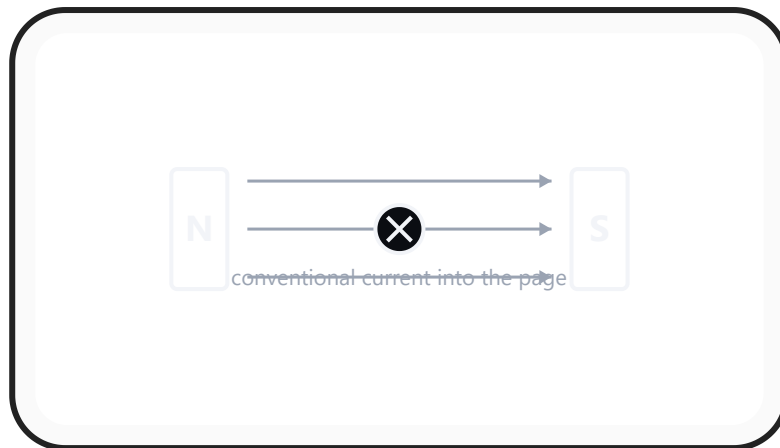


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=into_page,mode=assessment,pole_labels=N_S_horizontal,show_force_arrow=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

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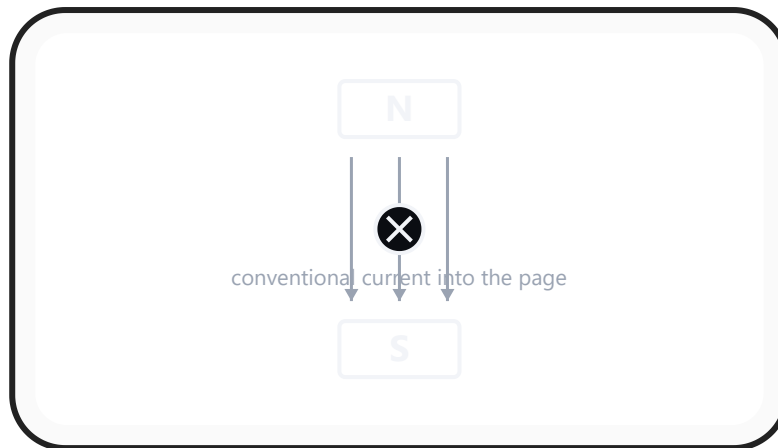


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=into_page,mode=assessment,pole_labels=N_S_vertical,show_force_arrow=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

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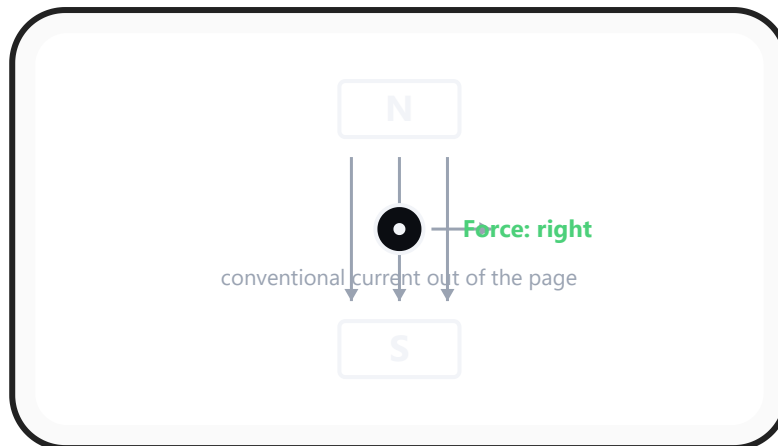


Diagram blueprint

Visual semantic contract

Canonical variant id

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=out_of_page,force_direction=right,mode=teaching,pole_labels=N_S_vertical,show_force_arrow=true::teaching

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

electrical.magnetism_and_electromagnetism

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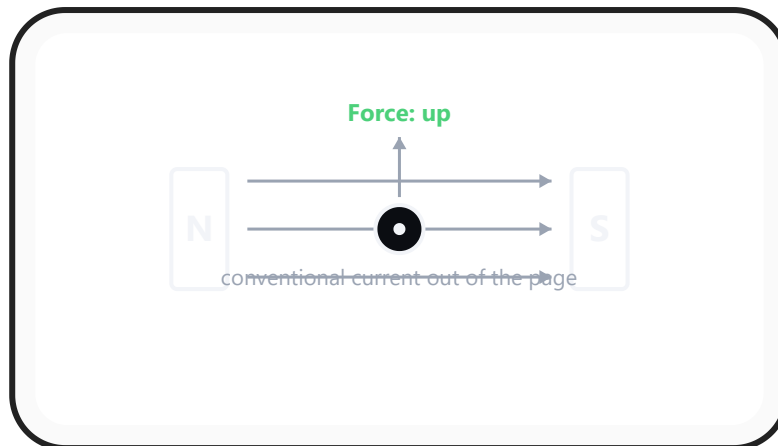


Diagram blueprint

Visual semantic contract

Canonical variant id

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=out_of_page,force_direction=up,mode=teaching,pole_labels=N_S_horizontal,show_force_arrow=true::teaching

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

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- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

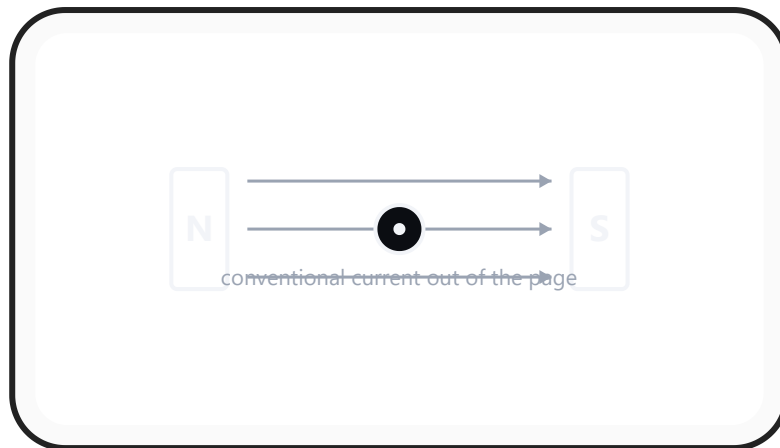


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=out_of_page,mode=assessment,pole_labels=N_S_horizontal,show_force_arrow=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

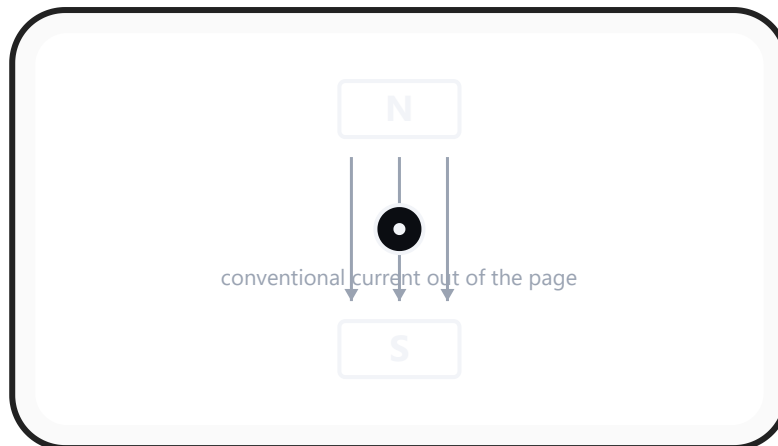


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=out_of_page,mode=assessment,pole_labels=N_S_vertical,show_force_arrow=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Show resistors connected as independent branches between two shared rails.

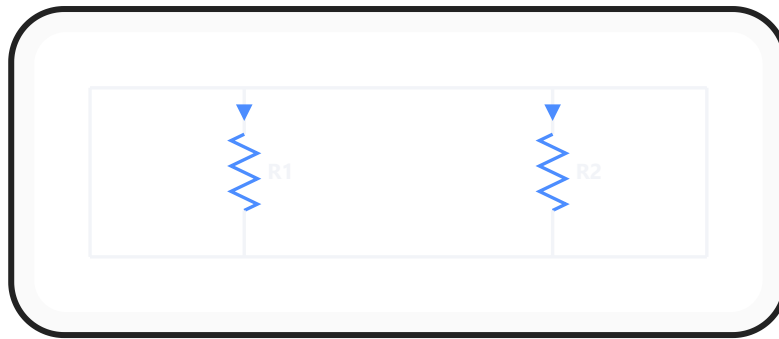


Diagram blueprint

circuit.parallel_resistors (electrical_circuit)

Visual semantic contract

visual-contract.parallel-circuit-branches

Canonical variant id

visual-contract.parallel-circuit-branches@1::branch_count=2,mode=both,show_branch_current_arrows=true,show_values=false::both

Linked assertion family

electrical.parallel_circuits

Linked assertion identifiers

EL-PARALLEL-RESISTANCE-CALC-001, EL-PARALLEL-RESISTANCE-001, EL-CIRCUIT-PARALLEL-STRUCTURE-001, EL-PARALLEL-CURRENT-CALC-001, EL-PARALLEL-DOMINANT-RESISTOR-001

Linked capability

cap.parallel.calculate_total_resistance, cap.parallel.solve_missing_branch, cap.parallel.recognise_structure, cap.parallel.calculate_branch_current, cap.parallel.identify_dominant_branch

Representation role

essential

MUST SHOW

- two shared horizontal rails
- every branch drawn as a separate vertical connection between the rails, labelled R1..Rn

MUST NOT SHOW

- branches drawn as a single series loop
- numeric component values

Show resistors connected as independent branches between two shared rails.



Diagram blueprint

circuit.parallel_resistors (electrical_circuit)

Visual semantic contract

visual-contract.parallel-circuit-branches

Canonical variant id

visual-contract.parallel-circuit-branches@1::branch_count=3,mode=both,show_branch_current_arrows=true,show_values=false::both

Linked assertion family

electrical.parallel_circuits

Linked assertion identifiers

EL-PARALLEL-RESISTANCE-CALC-001, EL-PARALLEL-RESISTANCE-001, EL-CIRCUIT-PARALLEL-STRUCTURE-001, EL-PARALLEL-CURRENT-CALC-001, EL-PARALLEL-DOMINANT-RESISTOR-001

Linked capability

cap.parallel.calculate_total_resistance, cap.parallel.solve_missing_branch, cap.parallel.recognise_structure, cap.parallel.calculate_branch_current, cap.parallel.identify_dominant_branch

Representation role

essential

MUST SHOW

- two shared horizontal rails
- every branch drawn as a separate vertical connection between the rails, labelled R1..Rn

MUST NOT SHOW

- branches drawn as a single series loop
- numeric component values

Show resistors connected as independent branches between two shared rails.

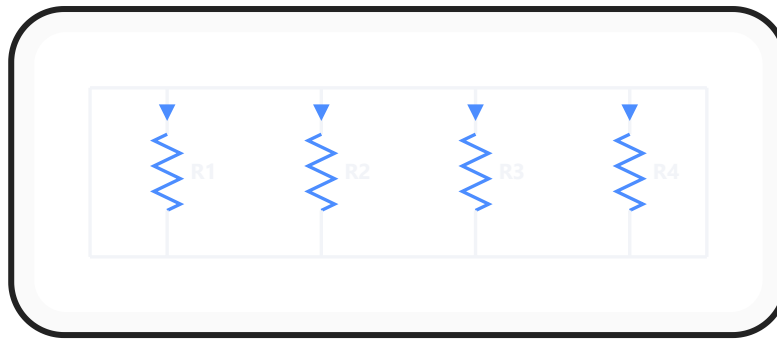


Diagram blueprint

circuit.parallel_resistors (electrical_circuit)

Visual semantic contract

visual-contract.parallel-circuit-branches

Canonical variant id

visual-contract.parallel-circuit-branches@1::branch_count=4,mode=both,show_branch_current_arrows=true,show_values=false::both

Linked assertion family

electrical.parallel_circuits

Linked assertion identifiers

EL-PARALLEL-RESISTANCE-CALC-001, EL-PARALLEL-RESISTANCE-001, EL-CIRCUIT-PARALLEL-STRUCTURE-001, EL-PARALLEL-CURRENT-CALC-001, EL-PARALLEL-DOMINANT-RESISTOR-001

Linked capability

cap.parallel.calculate_total_resistance, cap.parallel.solve_missing_branch, cap.parallel.recognise_structure, cap.parallel.calculate_branch_current, cap.parallel.identify_dominant_branch

Representation role

essential

MUST SHOW

- two shared horizontal rails
- every branch drawn as a separate vertical connection between the rails, labelled R1..Rn

MUST NOT SHOW

- branches drawn as a single series loop
- numeric component values

Teach the right-hand grip rule for the magnetic field circulating around a current-carrying conductor.

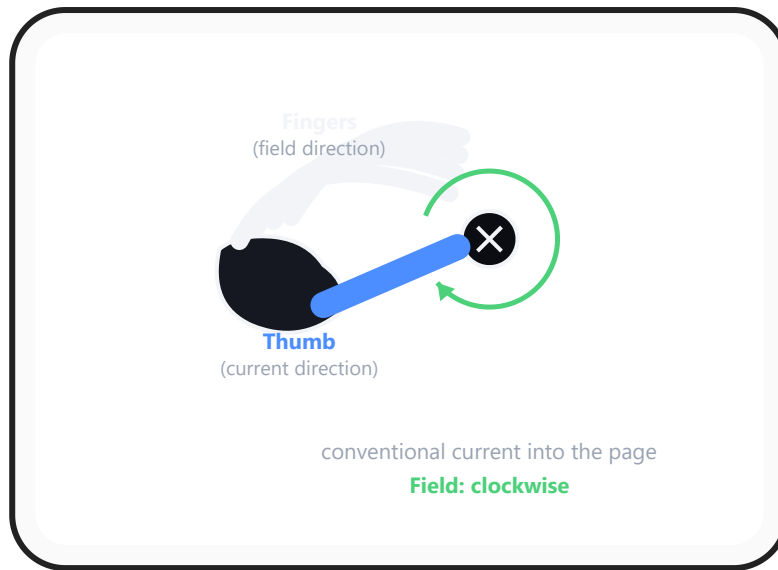


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.field_conductor_direction (magnetic_field)

visual-contract.right-hand-grip-rule

visual-contract.right-hand-grip-rule@1::current_direction=into_page,field_rotation=clockwise,mode=teaching,show_field_arrows=true::teaching

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FIELD-CURRENT-001

cap.magnetism.interpret_field_direction

essential

MUST SHOW

- a recognisable right hand
- a distinctly labelled thumb
- distinctly labelled curled fingers
- the conductor cross-section the hand grips

MUST NOT SHOW

- a generic force arrow presented as if it were the field direction (the pre-CC-05C-DIAGRAM-FIX defect: a generic magnetic-force diagram shown for a right-hand-grip-rule lesson)
- the field-rotation arrow/label in assessment mode (it is the assessed answer)

Teach the right-hand grip rule for the magnetic field circulating around a current-carrying conductor.

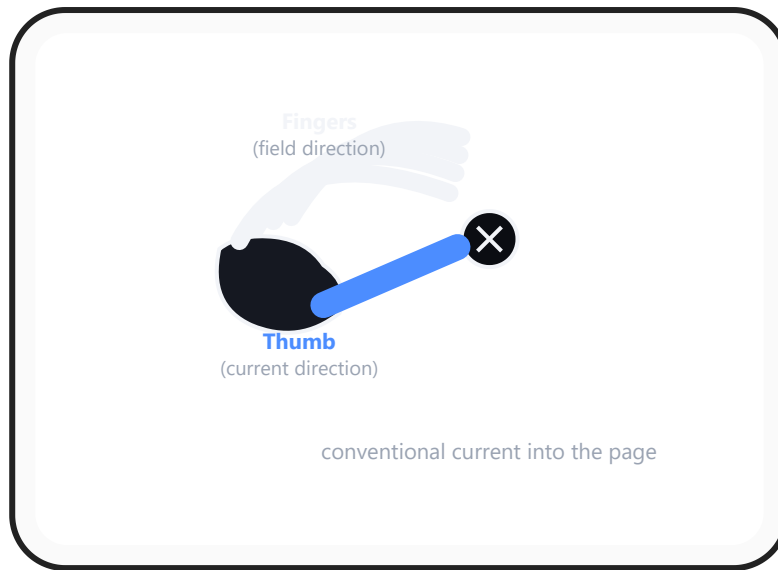


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.field_conductor_direction (magnetic_field)

visual-contract.right-hand-grip-rule

visual-contract.right-hand-grip-rule@1::current_direction=into_page,mode=assessment,show_field_arrows=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FIELD-CURRENT-001

cap.magnetism.interpret_field_direction

essential

MUST SHOW

- a recognisable right hand
- a distinctly labelled thumb
- distinctly labelled curled fingers
- the conductor cross-section the hand grips

MUST NOT SHOW

- a generic force arrow presented as if it were the field direction (the pre-CC-05C-DIAGRAM-FIX defect: a generic magnetic-force diagram shown for a right-hand-grip-rule lesson)
- the field-rotation arrow/label in assessment mode (it is the assessed answer)

Teach the right-hand grip rule for the magnetic field circulating around a current-carrying conductor.

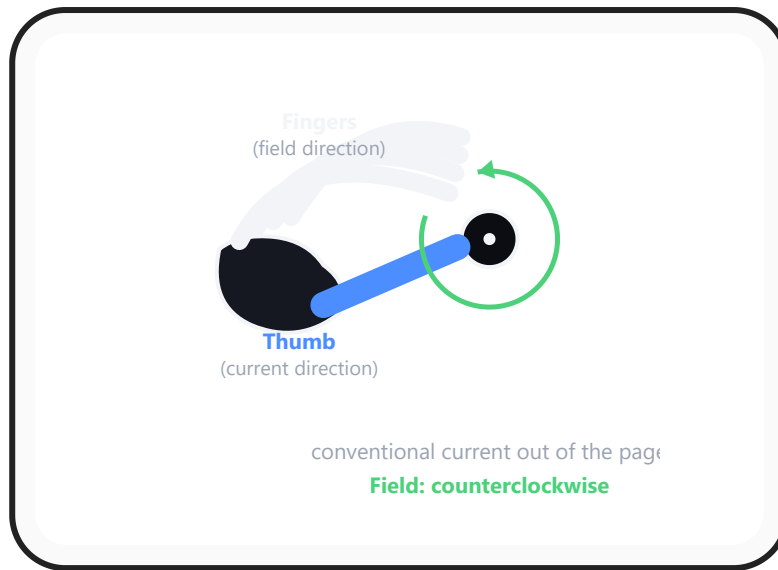


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.field_conductor_direction (magnetic_field)

visual-contract.right-hand-grip-rule

visual-contract.right-hand-grip-rule@1::current_direction=out_of_page,field_rotation=counterclockwise,mode=teaching,show_field_arrows=true::teaching

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FIELD-CURRENT-001

cap.magnetism.interpret_field_direction

essential

MUST SHOW

- a recognisable right hand
- a distinctly labelled thumb
- distinctly labelled curled fingers
- the conductor cross-section the hand grips

MUST NOT SHOW

- a generic force arrow presented as if it were the field direction (the pre-CC-05C-DIAGRAM-FIX defect: a generic magnetic-force diagram shown for a right-hand-grip-rule lesson)
- the field-rotation arrow/label in assessment mode (it is the assessed answer)

Teach the right-hand grip rule for the magnetic field circulating around a current-carrying conductor.

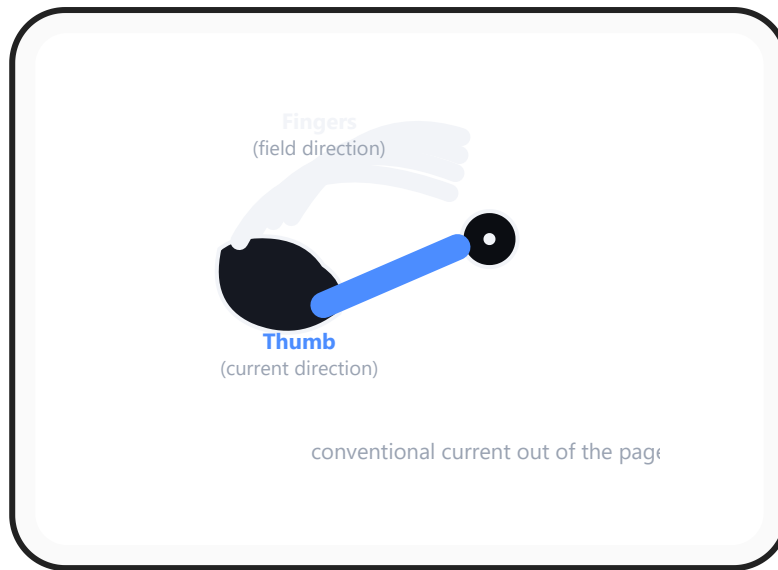


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.field_conductor_direction (magnetic_field)

visual-contract.right-hand-grip-rule

visual-contract.right-hand-grip-rule@1::current_direction=out_of_page,mode=assessment,show_field_arrows=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FIELD-CURRENT-001

cap.magnetism.interpret_field_direction

essential

MUST SHOW

- a recognisable right hand
- a distinctly labelled thumb
- distinctly labelled curled fingers
- the conductor cross-section the hand grips

MUST NOT SHOW

- a generic force arrow presented as if it were the field direction (the pre-CC-05C-DIAGRAM-FIX defect: a generic magnetic-force diagram shown for a right-hand-grip-rule lesson)
- the field-rotation arrow/label in assessment mode (it is the assessed answer)

Show resistors connected one after another in a single loop, with an arrow indicating the single shared current direction around the loop.

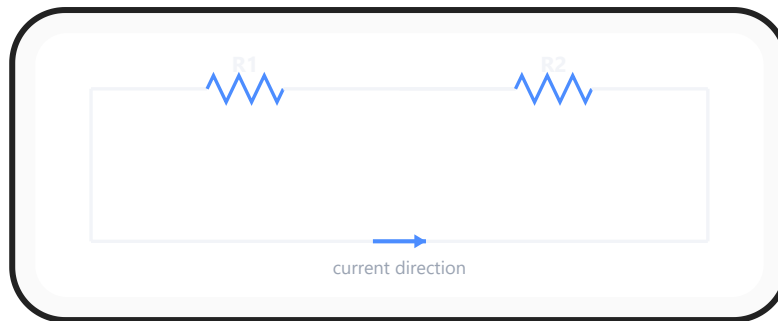


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

`circuit.series_resistors (electrical_circuit)``visual-contract.series-circuit-current-direction``visual-contract.series-circuit-current-direction@1::component_count=2,mode=both,show_current_arrow=true,show_values=false::both``electrical.series_circuits``EL-SERIES-RESISTANCE-CALC-001, EL-SERIES-RESISTANCE-001, EL-SERIES-VOLTAGE-CALC-001, EL-SERIES-DOMINANT-RESISTOR-001, EL-CIRCUIT-SERIES-STRUCTURE-001``cap.series.calculate_total_resistance, cap.series.solve_missing_component, cap.series.calculate_voltage_drop, cap.series.identify_dominant_component, cap.series.recognise_structure``essential`**MUST SHOW**

- one continuous single loop
- every resistor drawn in the loop, labelled R1..Rn in order
- a current-direction arrow on the loop, pointing along the wire it sits on

MUST NOT SHOW

- a second, parallel current path
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)

Show resistors connected one after another in a single loop, with an arrow indicating the single shared current direction around the loop.

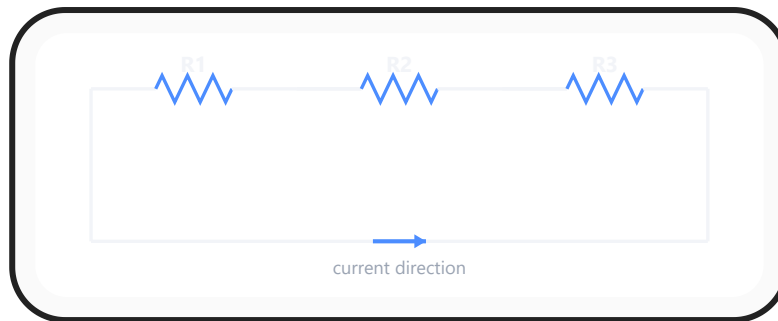


Diagram blueprint

`circuit.series_resistors (electrical_circuit)`

Visual semantic contract

`visual-contract.series-circuit-current-direction`

Canonical variant id

`visual-contract.series-circuit-current-direction@1::component_count=3,mode=both,show_current_arrow=true,show_values=false::both`

Linked assertion family

`electrical.series_circuits`

Linked assertion identifiers

`EL-SERIES-RESISTANCE-CALC-001, EL-SERIES-RESISTANCE-001, EL-SERIES-VOLTAGE-CALC-001, EL-SERIES-DOMINANT-RESISTOR-001, EL-CIRCUIT-SERIES-STRUCTURE-001`

Linked capability

`cap.series.calculate_total_resistance, cap.series.solve_missing_component, cap.series.calculate_voltage_drop, cap.series.identify_dominant_component, cap.series.recognise_structure`

Representation role

`essential`

MUST SHOW

- one continuous single loop
- every resistor drawn in the loop, labelled R1..Rn in order
- a current-direction arrow on the loop, pointing along the wire it sits on

MUST NOT SHOW

- a second, parallel current path
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)

Show resistors connected one after another in a single loop, with an arrow indicating the single shared current direction around the loop.

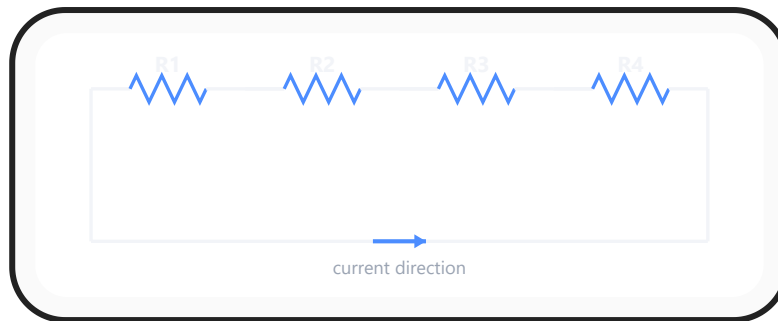


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

`circuit.series_resistors (electrical_circuit)`

`visual-contract.series-circuit-current-direction`

`visual-contract.series-circuit-current-direction@1::component_count=4,mode=both,show_current_arrow=true,show_values=false::both`

`electrical.series_circuits`

EL-SERIES-RESISTANCE-CALC-001, EL-SERIES-RESISTANCE-001, EL-SERIES-VOLTAGE-CALC-001, EL-SERIES-DOMINANT-RESISTOR-001, EL-CIRCUIT-SERIES-STRUCTURE-001

`cap.series.calculate_total_resistance`, `cap.series.solve_missing_component`, `cap.series.calculate_voltage_drop`, `cap.series.identify_dominant_component`, `cap.series.recognise_structure`

essential

MUST SHOW

- one continuous single loop
- every resistor drawn in the loop, labelled R1..Rn in order
- a current-direction arrow on the loop, pointing along the wire it sits on

MUST NOT SHOW

- a second, parallel current path
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)

Show a genuinely mixed series-and-parallel topology so learners can distinguish it from pure series or pure parallel, and trace the current path(s) through it.



Diagram blueprint	circuit.series_parallel_mixed (electrical_circuit)
Visual semantic contract	visual-contract.series-parallel-mixed-topology
Canonical variant id	visual-contract.series-parallel-mixed-topology@1::branch_arrangement=parallel_of_series,mode=both,show_values=false::both
Linked assertion family	electrical.series_vs_parallel_comparison
Linked assertion identifiers	EL-CIRCUIT-SELECT-CONFIGURATION-001, EL-CIRCUIT-MIXED-SERIES-PARALLEL-RECOGNITION-001, EL-CIRCUIT-TRACE-CURRENT-PATH-001
Linked capability	cap.comparison.identify_topology, cap.comparison.recognise_mixed_circuit, cap.comparison.trace_current_path
Representation role	essential

MUST SHOW

- a genuine series run (one or more components in a single current path)
- a genuine parallel group (two or more components between shared local rails) within the same image
- symbolic R1..Rn labels in a single consistent reading order
- the specific arrangement (series-of-parallel vs parallel-of-series) visually distinguishable from the other

MUST NOT SHOW

- a topology that is actually pure series or pure parallel dressed up to look mixed
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)

Show a genuinely mixed series-and-parallel topology so learners can distinguish it from pure series or pure parallel, and trace the current path(s) through it.



Diagram blueprint	circuit.series_parallel_mixed (electrical_circuit)
Visual semantic contract	visual-contract.series-parallel-mixed-topology
Canonical variant id	visual-contract.series-parallel-mixed-topology@1::branch_arrangement=series_of_parallel,mode=both,show_values=false::both
Linked assertion family	electrical.series_vs_parallel_comparison
Linked assertion identifiers	EL-CIRCUIT-SELECT-CONFIGURATION-001, EL-CIRCUIT-MIXED-SERIES-PARALLEL-RECOGNITION-001, EL-CIRCUIT-TRACE-CURRENT-PATH-001
Linked capability	cap.comparison.identify_topology, cap.comparison.recognise_mixed_circuit, cap.comparison.trace_current_path
Representation role	essential

MUST SHOW

- a genuine series run (one or more components in a single current path)
- a genuine parallel group (two or more components between shared local rails) within the same image
- symbolic R1..Rn labels in a single consistent reading order
- the specific arrangement (series-of-parallel vs parallel-of-series) visually distinguishable from the other

MUST NOT SHOW

- a topology that is actually pure series or pure parallel dressed up to look mixed
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)